

Predicting Medical Costs and Motivation Dimension Reduction: An Application of Optimal Scaling Methods

Eulàlia Gómez-Aguiló
Taymaz Ghaneh
Alexandra Constantin



Overview

1. Optimal Scaling Regression

- Scaling level choice
- Best model
- Transformation Plots

1. Optimal Scaling PCA

- PCA vs OS-PCA
- Dimensionality
- Item groups

Optimal Scaling Regression

OSR - Modeling

Model 1

- charges - numeric
- age - numeric
- sex - numeric
- BMI - numeric
- smoker - numeric
- region - nominal

Default discretisation for *charges*

OSR - Modeling

Model 2

- charges - numeric
- **age - spline-nominal** (degree=2, knots=6)
- sex - numeric
- **BMI - spline-nominal** (degree=2, knots=6)
- smoker - numeric
- region - nominal

Grouping discretisation for *charges*

OSR - Modelling

Model 3

- charges - numeric
- age - spline-nominal (degree=2, knots=6)
- sex - numeric
- **BMI - spline-ordinal** (degree=2, knots=6)
- smoker - numeric
- region - nominal

Grouping discretisation for *charges*

OSR - Results

Model	R²	Adj-R²	APE	EPE
<i>Model 1</i>	.744	.740	.256	.269
<i>Model 2</i>	.760	.746	.240	.272
<i>Model 3</i>	.757	.747	.243	.268

OSR - Results

Model	R ²	Adj-R ²	APE	EPE
<i>Model 1</i>	.744	.740	.256	.269
 <i>Model 2</i>	.760	.746	.240	.272
<i>Model 3</i>	.757	.747	.243	.268

OSR - Coefficients

Predictor	β	SE	F	p
Age of primary beneficiary	.322	.026	147.34	< .001
Sex	.018	.026	.457	.499
Body mass index (BMI)	.182	.026	48.15	< .001
Smoking status	.778	.023	1114.74	< .001
Residential area	.026	.019	1.882	.132

OSR - Coefficients

Predictor	β	SE	F	p
Age of primary beneficiary	.322	.026	147.34	< .001
Sex	.018	.026	.457	.499
Body mass index (BMI)	.182	.026	48.15	< .001
Smoking status	.778	.023	1114.74	< .001
Residential area	.026	.019	1.882	.132

OSR - Coefficients

Predictor	β	SE	F	p
Age of primary beneficiary	.322	.026	147.34	< .001
Sex	.018	.026	.457	.499
Body mass index (BMI)	.182	.026	48.15	< .001
Smoking status	.778	.023	1114.74	< .001
Residential area	.026	.019	1.882	.132

OSR - Coefficients

Predictor	β	SE	F	p
Age of primary beneficiary	.322	.026	147.34	< .001
Sex	.018	.026	.457	.499
Body mass index (BMI)	.182	.026	48.15	< .001
Smoking status	.778	.023	1114.74	< .001
Residential area	.026	.019	1.882	.132

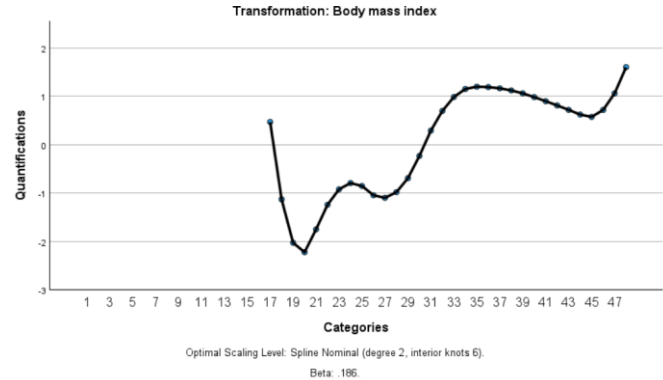
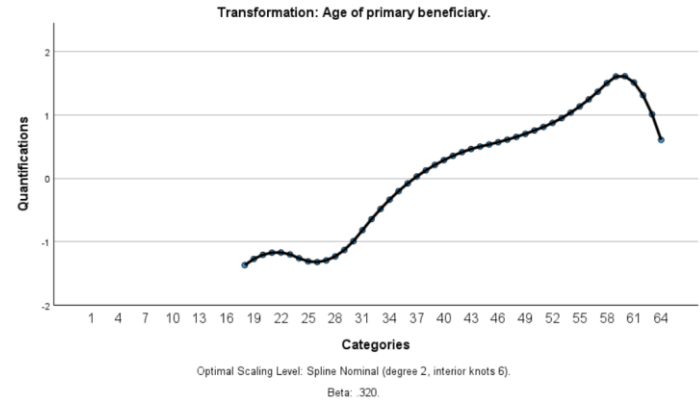
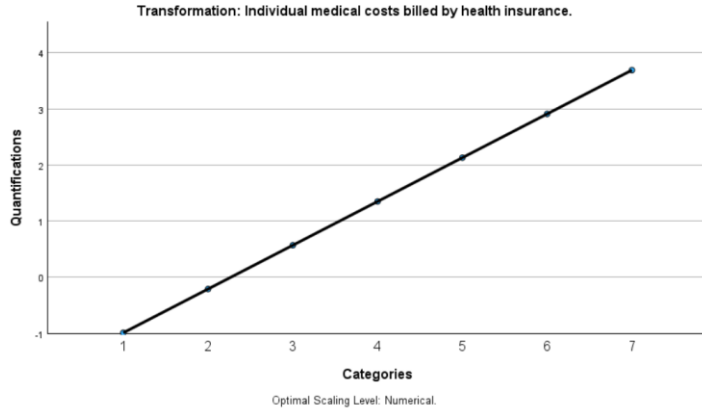
OSR - Coefficients

Predictor	Importance	Tolerance (after)	Tolerance (before)
Age of primary beneficiary	.129	.996	.991
Sex	.002	.991	.998
Body mass index (BMI)	.061	.972	.956
Smoking status	.805	.983	.987
Residential area	.003	.978	.953

OSR - Coefficients

Predictor	Importance	Tolerance (after)	Tolerance (before)
Age of primary beneficiary	.129	.996	.991
Sex	.002	.991	.998
Body mass index (BMI)	.061	.972	.956
Smoking status	.805	.983	.987
Residential area	.003	.978	.953

OSR - Transformation Plots

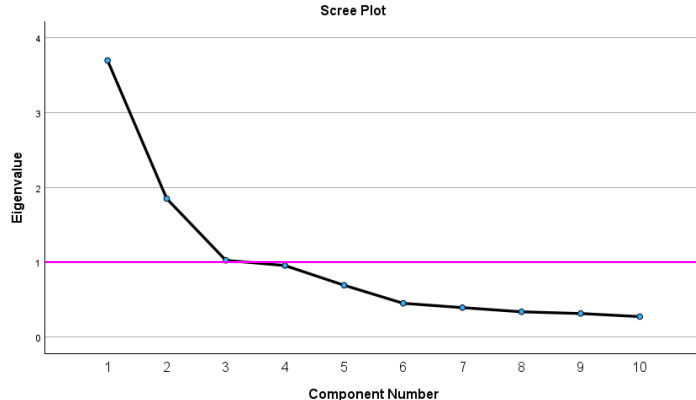


Optimal Scaling PCA

Linear PCA

10 items

(36.98 + 18.51 + 10.26) % :
65.76% CumVAF by 3 PCs
55.50 % CumVAF by 2 PCs



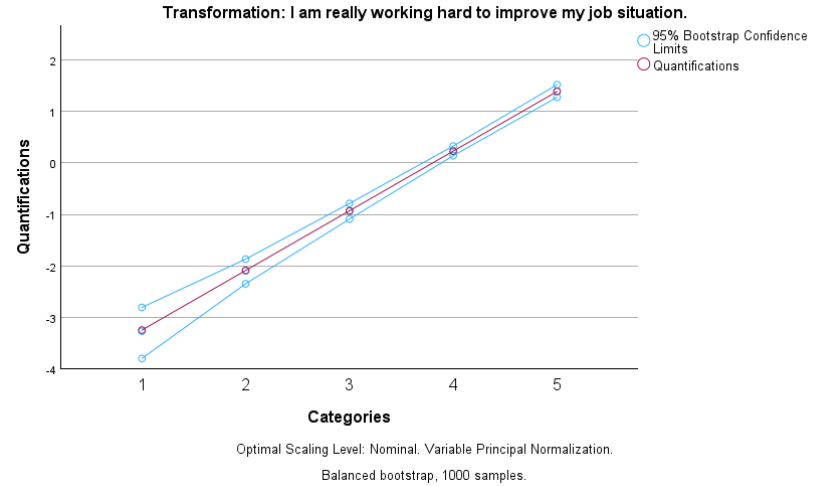
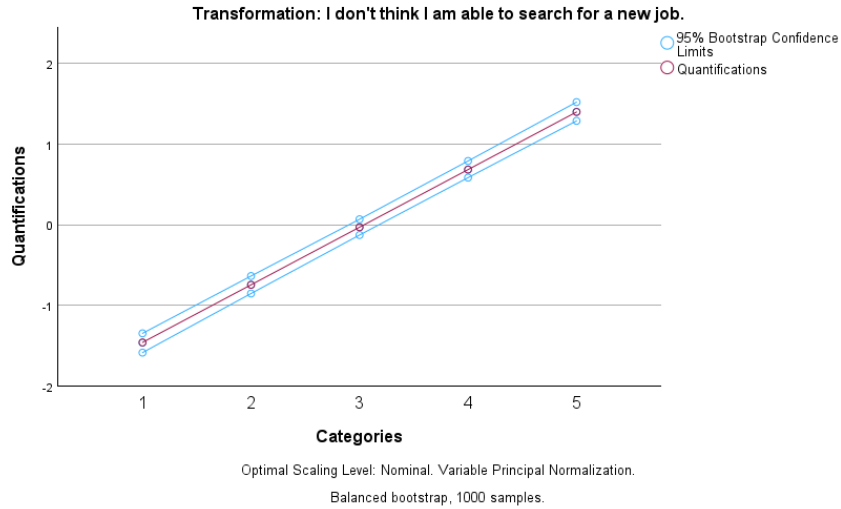
OS PCA

10 items

(36.99 + 18.51 + 10.26) % :
65.77% CumVAF by 3 PCs
55.51 % CumVAF by 2 PCs

Very similar: then why prefer OS PCA ?

- Items are **ordinal** (Likert-type), not continuous → OSPCA handles **freedom and constraints**
- OSPCA **searches** for quantifications that improve component **reliability** via Cronbach's alpha.
- OSPCA handles **non-linearities** in ordinal data and recovers more **interpretable** structure.

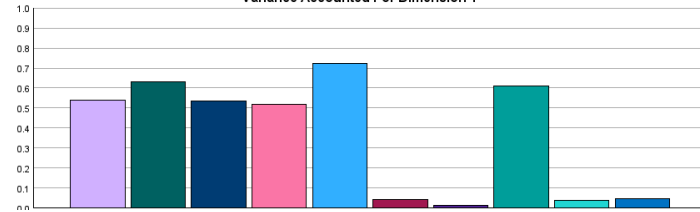


All 10 transformation plots are more or less monotonic and linear

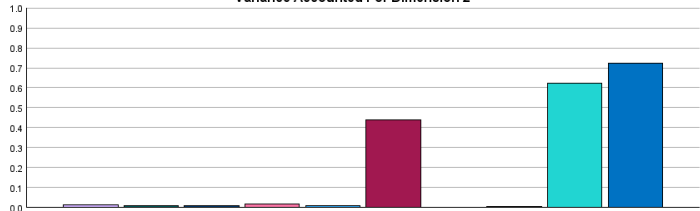
PC1	$\alpha \approx 0.8$: Good	VAF% ≈ 37 % Strong
PC2	$\alpha \approx 0.5$: Poor	VAF% ≈ 18.5 % Poor
PC3	$\alpha \approx 0.03$: Unacceptable	VAF% ≈ 10 % Just 1 of 10

- Need to correct the dimensionality and re-run the model.
- Should we also remove **Item 7** from the analysis ?? (Next Page)

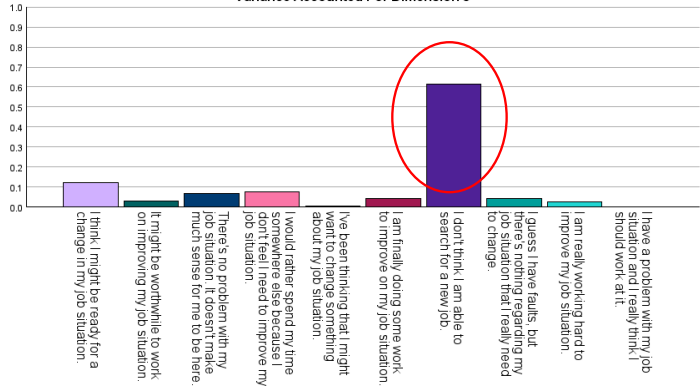
Variance Accounted For Dimension 1



Variance Accounted For Dimension 2



Variance Accounted For Dimension 3



Item 7 is heavily loaded on PC3

- Need to correct the dimensionality and re-run the model.
- Should we also remove **Item 7** from the analysis ??

Still 10 items, But now 2 PCs

- No sign flip in intervals : $0 \notin [L,U]$
- No major change during bootstrap samplings
- PC1 and PC2 :
 - ◆ Reliable (internally consistent)
 - ◆ Explanatory Strong : $\approx 60\%$

Model Summary

Dimension	Cronbach's Alpha	Variance Accounted For Total (Eigenvalue)	% of Variance
1	.840	4.100	41.001
2	.527	1.903	19.026
Total	.926 ^a	6.003	60.027

a. Total Cronbach's Alpha is based on the total Eigenvalue.

No major change

Estimates are in the CL range

Model Summary Bootstrap Results^a

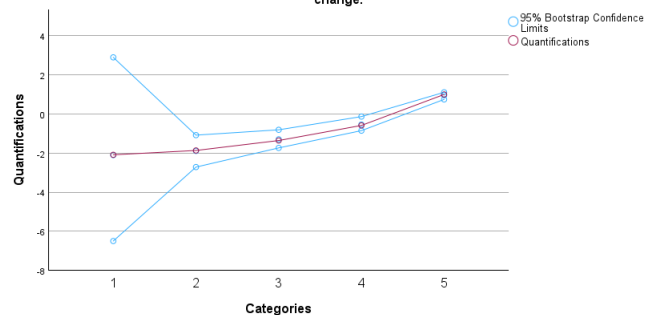
Dimension	Cronbach's Alpha			Variance Accounted For					
	95% Bootstrap Confidence Limits			Total (Eigenvalue)			% of Variance		
	Bootstrap Mean	Lower	Upper	Bootstrap Mean	Lower	Upper	Bootstrap Mean	Lower	Upper
1	.843	.818	.862	4.143	3.791	4.463	41.426	37.911	44.629
2	.545	.471	.608	1.964	1.735	2.210	19.637	17.347	22.104
Total	.929	.920	.937	6.106	5.826	6.395	61.063	58.260	63.947

a. Balanced bootstrap (procustes), 1000 samples.

Still 10 items, But now 2 PCs

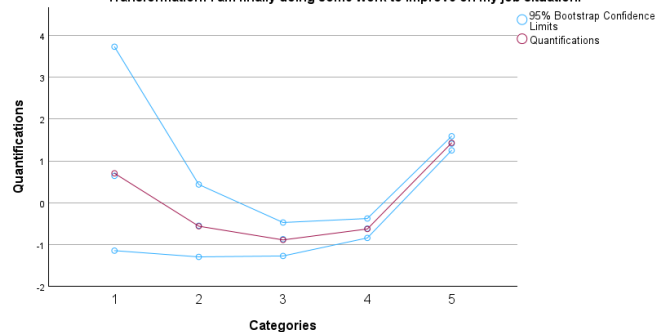
Item 7 : order not preserved under quantification

Transformation: I guess I have faults, but there's nothing regarding my job situation that I really need to change.



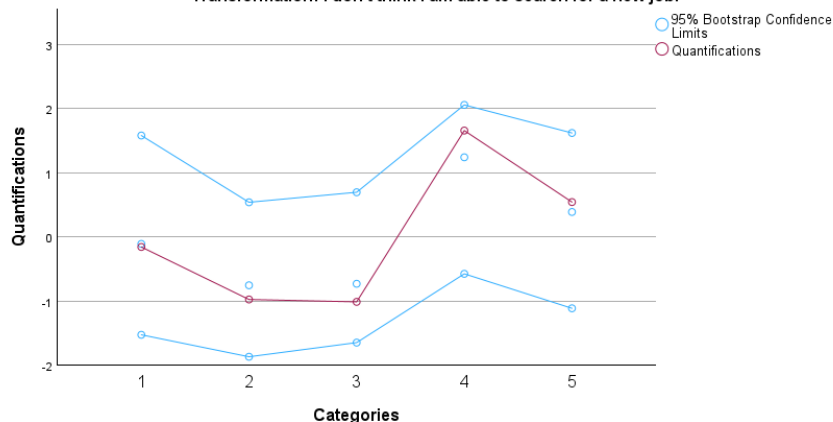
Optimal Scaling Level: Nominal. Variable Principal Normalization.
Balanced bootstrap, 1000 samples.

Transformation: I am finally doing some work to improve on my job situation.



Optimal Scaling Level: Nominal. Variable Principal Normalization.
Balanced bootstrap, 1000 samples.

Transformation: I don't think I am able to search for a new job.



Optimal Scaling Level: Nominal. Variable Principal Normalization.
Balanced bootstrap, 1000 samples.

When **constrain the number of components**, the algorithm must find transformations that *maximize variance explained by ONLY those selected components* (Gifi,1990): Hence **Non-monotonic** transformations happen !

Item 7 : Unstable Results

I don't think I am able to search for a new job.^a

Category	Frequency	Quantification	Centroid Coordinates Dimension		Vector Coordinates Dimension	
			1	2	1	2
Strongly Disagree	75	-.156	-.082	.002	-.019	-.034
Disagree	68	-.974	-.067	-.243	-.121	-.212
Undecided	89	-1.011	-.136	-.214	-.125	-.220
Agree	75	1.658	.215	.356	.205	.362
Strongly Agree	80	.544	.083	.110	.067	.119

Variable Principal Normalization.

a. Optimal Scaling Level: Nominal.

Item 7 when 2 PCs

(Dimension Reduction Applied)

I don't think I am able to search for a new job. Bootstrap Results^{a,b}

Category	Frequency	Quantification			Vector Coordinates							Ellipse Area	
		Bootstrap Mean	95% Bootstrap Confidence Limits		Bootstrap Mean	Dimension 1			Bootstrap Mean	Dimension 2			
			Lower	Upper		95% Bootstrap Confidence Limits				95% Bootstrap Confidence Limits			
						Lower	Upper	Lower		Upper	Lower		Upper
Strongly Disagree	75	-.108	-1.524	1.581	-.050	-.318	.140	-.006	-.372	.420	.506		
Disagree	68	-.752	-1.866	.541	-.077	-.306	.191	-.214	-.621	.106	.492		
Undecided	89	-.729	-1.645	.697	-.107	-.382	.093	-.198	-.534	.152	.408		
Agree	75	1.241	-.574	2.057	.179	-.084	.485	.327	-.133	.734	.617		
Strongly Agree	80	.389	-1.111	1.620	.060	-.143	.288	.102	-.333	.524	.441		

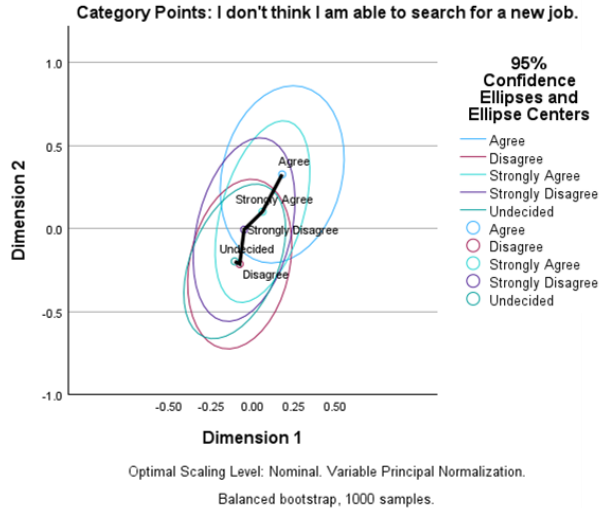
Variable Principal Normalization.

a. Optimal Scaling Level: Nominal.

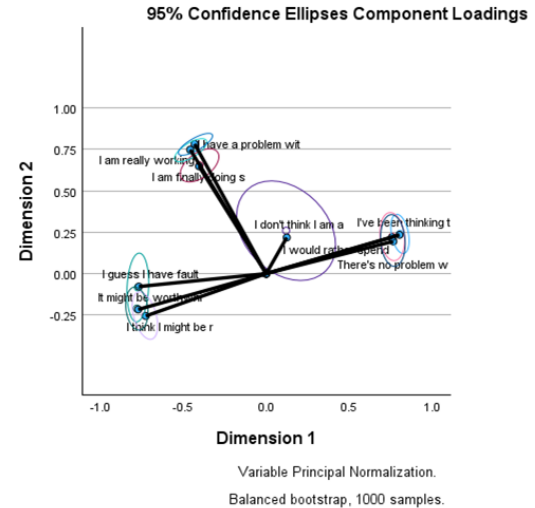
b. Balanced bootstrap (procrustes), 1000 samples.

Sign flip

Item 7 : Unstable Results



- ◆ Not ordered
- ◆ Unstable alignment
- ◆ Crosses quadrants borders



- ◆ Small Communality
- ◆ Unstable Loading vector

Item 7 : Unstable Results



- Shall we remove **Item 7** now ??

Item 7 : Unstable results

A matter of Quantification

Or ?

A matter of Dimension Reduction

Item 7 when 10 PCs

(before Dimension Reduction)

I don't think I am able to search for a new job: Bootstrap Results^{a,b}

Category	Frequency	Quantification			Vector Coordinates								
		Bootstrap Mean	95% Bootstrap Confidence Limits		Bootstrap Mean	95% Bootstrap Confidence Limits		Bootstrap Mean	95% Bootstrap Confidence Limits		Bootstrap Mean	95% Bootstrap Confidence Limits	
			Lower	Upper		Lower	Upper		Lower	Upper		Lower	Upper
Strongly Disagree	75	-1.460	-1.583	-1.344	-.162	-.251	-.053	-.013	-.117	.084	-1.144	-1.252	-1.040
Disagree	68	-.743	-.852	-.635	-.083	-.132	-.027	-.006	-.058	.044	-.582	-.672	-.501
Undecided	89	-.031	-.128	.067	-.003	-.015	.008	.000	-.005	.003	-.025	-.100	.051
Agree	75	.682	.582	.788	.076	.024	.123	.006	-.039	.052	.534	.457	.619
Strongly Agree	80	1.397	1.283	1.516	.155	.050	.244	.012	-.078	.109	1.093	1.007	1.187

Variable Principal Normalization.

a. Optimal Scaling Level: Nominal.

b. Balanced bootstrap (procrustes), 1000 samples.

cropped

9 items : *Doei item 7*

Iteration History					
Iteration Number	Variance Accounted For		Total	Loss	
	Total	Increase		Centroid Coordinates	Restriction of Centroid to Vector Coordinates
0 ^a	5.540815	.000005	12.459185	12.116916	.342269
24 ^b	5.968551	.000008	12.031449	11.766720	.264729

a. Iteration 0 displays the statistics of the solution with all variables, except variables with optimal scaling level Multiple Nominal, treated as numerical.

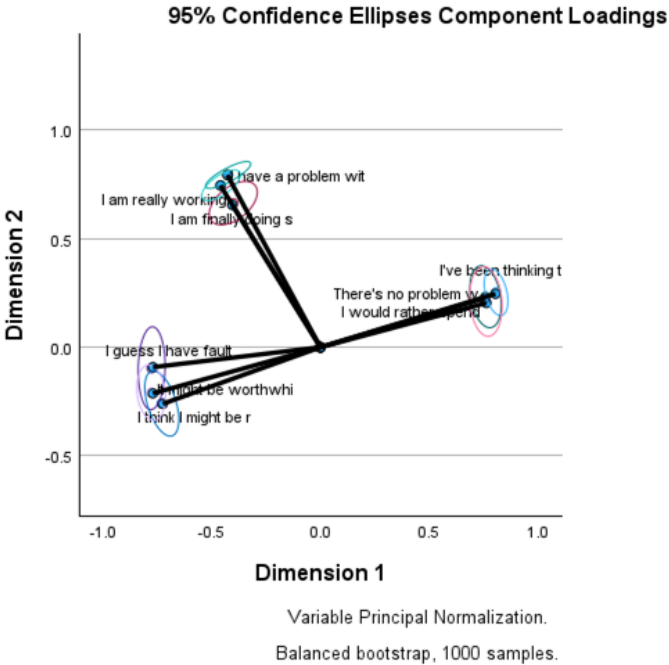
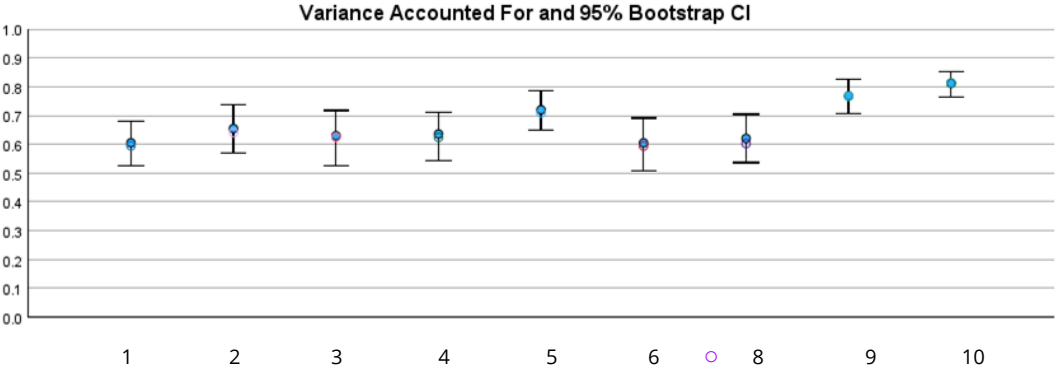
b. The iteration process stopped because the convergence test value was reached.

Empirical Centroids have got closer to category-loading vectors → better fit, smaller loss

PC1	$\alpha \approx 0.85$: Good	VAF% ≈ 41 % +5% improvement*
PC2	$\alpha \approx 0.53$: Poor	VAF% ≈ 19 % +0.5% improvement*

* improvement due to item 7 removal

9 items - 2 PCs

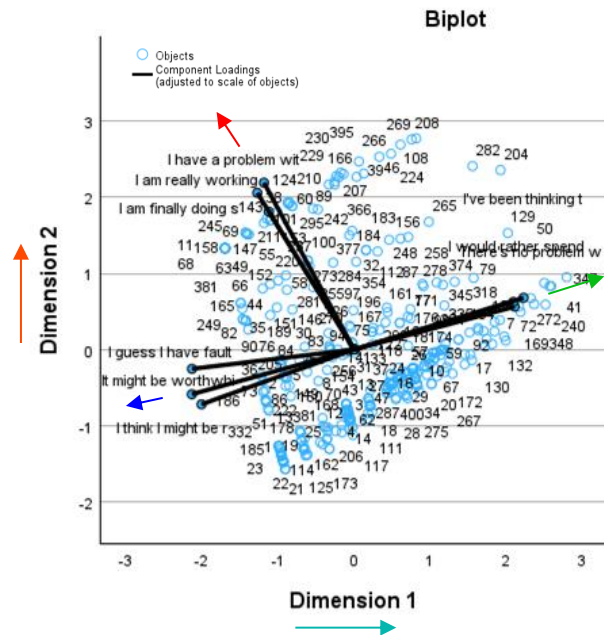


Latent factor interpretation

- Dimension 1: Self-awareness / perceived need for change
- Dimension 2: Action orientation / behavioural engagement

- ◆ Pre-contemplation (No intention / Denial)
- ◆ Contemplation Phase (Ambivalence / Thinking about change)
- ◆ Action (Active change / Improvement behavior)

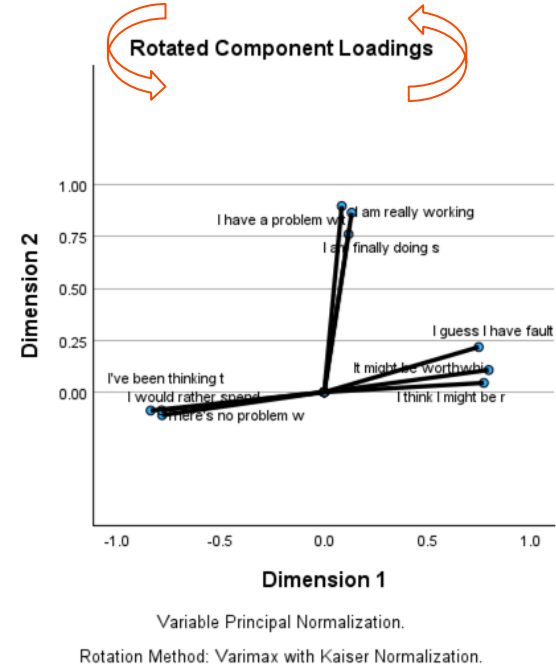
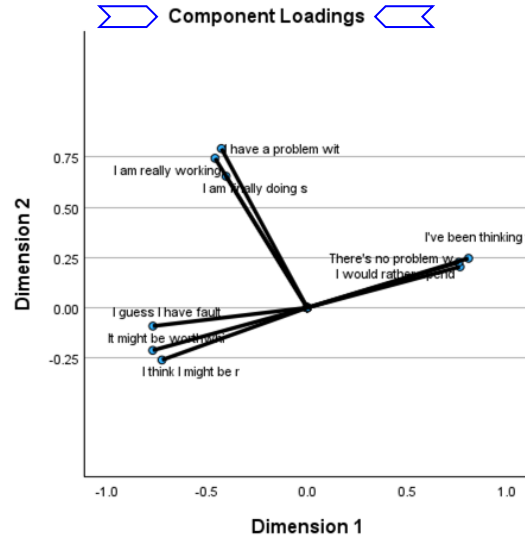
But PCA is **NOT** a Clustering Method



To Rotate

Or

Not to Rotate



- No improvement in VAFs by **Varimax** rotation
- But items now better aligned with one of the two PCs → more **interpretability**

Summary

Summary

OSR

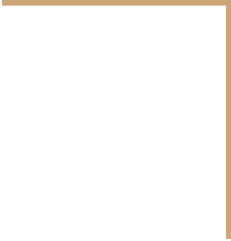
Model 2 ($R^2 = .760$; EPE = .272)

- charges - numeric
- **smoker** - numeric
- **age** - spline-nominal (degree=2, knots=6)
- **BMI** - spline-nominal (degree=2, knots=6)
- sex - numeric
- region - nominal

OS-PCA

- Dimensions:
 - **Dimension 1**: Self-awareness / perceived need for change
 - **Dimension 2**: Action orientation / behavioural engagement
- Item groups:
 - **Action Phase** (Motivation 6, 9, 10)
 - **Contemplation** (Motivation 1, 2, 8)
 - **Pre-contemplation** (Motivation 3, 4, 5)

Thank you!



Questions?

